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DTE Energy

10 CFR 50.73

December 20, 2016
NRC-16-0072

U. S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, D.C. 20555-0001

Reference: Fermi 2
NRC Docket No. 50-341
NRC License No. NPF-43

Subject: Licensee Event Report (LER) No. 2016-011

Pursuant to 10 CFR 50.73(a)(2)(i)(B) and 10 50.73(a)(2)(v)(A), (C), and (D), DTE Electric Company (DTE) is submitting LER No. 2016-011, Standby Liquid Control Inoperable due to Sodium Pentaborate Concentration Outside of Technical Specifications.

No new commitments are being made in this LER.

Should you have any questions or require additional information, please contact Mr. Scott A. Maglio, Manager –Nuclear Licensing, at (734) 586-5076.

Sincerely,

Keith J. Polson
Site Vice President

Enclosure: Licensee Event Report No. 2016-011, Standby Liquid Control Inoperable due to Sodium Pentaborate Concentration Outside of Technical Specifications

cc: NRC Project Manager
NRC Resident Office
Reactor Projects Chief, Branch 5, Region III
Regional Administrator, Region III
Michigan Public Service Commission
Regulated Energy Division (kindschl@michigan.gov)

Enclosure to
NRC-16-0072

Fermi 2 NRC Docket No. 50-341
Operating License No. NPF-43

Licensee Event Report (LER) No. 2016-011, Standby Liquid Control Inoperable due to Sodium Pentaborate Concentration Outside of Technical Specifications

NRC FORM 366 (06-2016) U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104 EXPIRES: 10/31/2018
LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)	Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. PAGE
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4. TITLE

Standby Liquid Control Inoperable due to Sodium Pentaborate Concentration Outside of Technical Specifications

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME: N/A	DOCKET NUMBER: 05000
10	28	2016	2016	011	00	12	20	2016	FACILITY NAME: N/A	DOCKET NUMBER: 05000

9. OPERATING MODE: 1

10. POWER LEVEL: 100

11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: *(Check all that apply)*

☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☑ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☑ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☑ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(vi)	☑ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ OTHER Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT	TELEPHONE NUMBER *(Include Area Code)*
Fermi 2 / Scott A. Maglio – Manager, Nuclear Licensing	(734) 586-5076

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
B	BR	V	R340	Y	N/A	N/A	N/A	N/A	N/A

14. SUPPLEMENTAL REPORT EXPECTED	15. EXPECTED SUBMISSION DATE	MONTH	DAY	YEAR
☐ YES *(If yes, complete 15. EXPECTED SUBMISSION DATE)* ☑ NO				

ABSTRACT *(Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)*

On October 28, 2016, at 1500 EDT, Fermi 2 Operators declared both subsystems of Standby Liquid Control (SLC) inoperable due to the sodium pentaborate concentration in the SLC storage tank being outside of the Technical Specification (TS) requirements. The sodium pentaborate concentration had been measured in response to the receipt of the SLC storage tank high level alarm. The demineralized water isolation valves that could cause a dilution of the tank were verified to be closed. Actions were initiated to add sodium pentaborate to the SLC storage tank to raise the concentration. SLC was declared operable at 1935 EDT following a measurement that verified that sodium pentaborate concentration had been restored to within the TS requirements. The cause of the event was determined to be leak-by of the demineralized water isolation valves that had caused an increase in SLC storage tank level and, therefore, a decrease in sodium pentaborate concentration. The leak-by had been occurring for some time and had gone undetected. The cause of the leak-by not being detected was determined to be inadequate system monitoring. Corrective actions were completed to replace the demineralized water isolation valves and revise the chemistry specification to clearly specify action levels for the tank level and sodium pentaborate concentration and provide margin to the TS limits. Corrective actions were also completed to revise the SLC system monitoring plan to incorporate trending of tank level and boron concentration. The safety significance of this event is very low and the event did not pose a threat to the health and safety of the public or plant personnel. There were no radiological releases associated with this event.

NARRATIVE

INITIAL PLANT CONDITIONS

Mode – 1
Reactor Power – 100 percent

There were no structures, systems, or components (SSCs) that were inoperable at the start of this event that contributed to this event.

DESCRIPTION OF THE EVENT

At 1051 EDT on October 28, 2016, while operating in MODE 1 at 100% power, the Fermi 2 Main Control Room (MCR) [[NA]] received an alarm for Standby Liquid Control (SLC) [[BR]] tank [[TK]] level. The MCR and local level indications [[LI]] both were above the high SLC storage tank level alarm [[LA]] setpoint. Operators verified SLC storage tank demineralized water isolation valves [[V]] were closed per the Alarm Response Procedure (ARP). The SLC storage tank was then manually measured to confirm level indication at 1214 EDT. The measured level was below the high SLC storage tank level alarm setpoint. A chemistry sample of the SLC storage tank to determine the sodium pentaborate concentration was requested. In order to ensure a representative sample of the SLC storage tank, sparging of the SLC storage tank was begun at 1340 EDT. In accordance with plant procedures, SLC is considered inoperable during the sparging process and, therefore, Technical Specification (TS) Limiting Condition for Operation (LCO) 3.1.7, "Standby Liquid Control (SLC) System," Condition B was entered for two SLC subsystems inoperable. Required Action B.1 to restore one SLC subsystem has a Completion Time of 8 hours. Sparging was completed at 1404 EDT (i.e. 24 minutes after entering LCO 3.1.7 Condition B) and the SLC system was declared OPERABLE.

At 1500 EDT, the results of the chemistry sample of the SLC storage tank were obtained and indicated that the sodium pentaborate concentration was 8.3%. Surveillance Requirement (SR) 3.1.7.5 requires the concentration to be within the limits of Figure 3.1.7-1. For the SLC storage tank level at the time of the event (i.e. near the high level alarm setpoint), the sodium pentaborate concentration was required to be in the range from 8.5% to 9.5% (weight percent). As a result, SLC was declared inoperable at 1500 EDT and LCO 3.1.7 Condition B was entered for two SLC subsystems inoperable. A second sample was taken and the results confirmed the previous value of 8.3%. Actions were initiated to add sodium pentaborate to the SLC storage tank to raise the concentration. Subsequent sparging and sampling determined that the sodium pentaborate concentration had been raised to 8.9%. Due to the sodium pentaborate concentration being restored to within the SR 3.1.7.5 requirements, SLC was declared OPERABLE at 1935 EDT (i.e. approximately 4.5 hours after entering LCO 3.1.7 Condition B) and the LCO was exited. The SLC storage tank level was then manually lowered to clear the level alarm per plant procedures. Local level indications and manual measurements confirmed the SLC storage tank level was below the high level alarm setpoint by 2216 EDT.

A subsequent review of the event determined that the SLC storage tank level had been slowly increasing over several months due to introduction of water through one or both demineralized water isolation valves that were closed but leaking by. On October 9, a SLC pump and valve operability test was performed, in accordance with SR 3.1.7.7. Following the successful completion of this test, which involved cycling one of the demineralized water isolation valves, the valve leak-by increased as a more substantial increase in SLC storage tank level occurred. This increased leak-by was not detected by Operators until the occurrence of the high level alarm on October 28 as described above. Since the water leaking by the valves was demineralized, dilution of the sodium pentaborate concentration was occurring throughout this time. The frequency of TS SR 3.1.7.5 to determine the sodium pentaborate concentration is monthly. The most recent

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performances of this SR prior to this event were on September 15 and October 13 and both resulted in an acceptable sodium pentaborate concentration of 8.7%. It is not possible to establish exactly when the sodium pentaborate decreased below the TS SR 3.1.7.5 limit of 8.5%. However, it is reasonable to assume the concentration decrease was approximately linear which would indicate that the TS SR 3.1.7.5 limit was not met for several days prior to the event on October 28. As described previously, TS LCO 3.1.7 Condition B Required Action B.1 has a Completion Time of 8 hours. If the Completion Time is not met, Condition C has a Required Action C.1 to be in MODE 3 in 12 hours. Since the sum of these two Completion Times is less than one day (i.e. 20 hours), it is reasonable to assume that the condition of the sodium pentaborate concentration not meeting SR 3.1.7.5 lasted longer than allowed by TS. This represents an operation or condition prohibited by the plant TS and is therefore reportable under Title 10 Code of Federal Regulations (10 CFR) 50.73(a)(2)(i)(B). There is no corresponding requirement for event notification to the NRC under 10 CFR 50.72.

The Fermi 2 Updated Final Safety Analysis Report (UFSAR) states that the SLC system provides a redundant, independent, and different way to bring the nuclear fission reactor to subcriticality and maintain subcriticality as the reactor cools. The SLC system permits an orderly and safe shutdown in the event that control rods cannot be inserted into the reactor core in sufficient number to accomplish shutdown in the normal manner. In addition, the SLC system is also credited for injecting sodium pentaborate into the reactor coolant system after a design basis loss of coolant accident (LOCA) in order to control the pH of Emergency Core Cooling System (ECCS) water to prevent iodine re-evolution. The SLC system consists of a SLC storage tank, two positive displacement pumps [[P]], two explosive valves, and associated piping and valves used to transfer the borated water from the storage tank to the reactor pressure vessel (RPV) [[RPV]]. Since there is only a single storage tank, conditions affecting the tank impact the operability of both SLC subsystems.

As described above, both SLC subsystems were declared inoperable due to the sodium pentaborate concentration being outside of the TS required range. This represents a condition that could have prevented the fulfillment of the safety functions of the SLC system described above. Since the conditions at the time of discovery on October 28 met the reporting criteria of 10 CFR 50.72(b)(3)(v)(A), (C), and (D) as an event or condition that could have prevented the fulfillment of a safety function needed to shut down the reactor and maintain it in a safe shutdown, control the release of radioactive material, or mitigate the consequences of an accident, an 8-hour non-emergency event notification (EN 52331) was made to the NRC. This LER 2016-011 is being reported under the corresponding requirements in 10 CFR 50.73(a)(2)(v)(A), (C), and (D).

SIGNIFICANT SAFETY CONSEQUENCES AND IMPLICATIONS

As described above, the SLC system performs two different functions: (1) enable safe shutdown in the event that control rods cannot be inserted into the reactor core and (2) lower the pH of ECCS water to prevent iodine re-evolution post-LOCA. The safety consequences and implications of each of these two functions are addressed below.

Regarding the first function, the SLC system is a redundant and independent means to control reactivity. A probabilistic risk assessment (PRA) was performed to determine the impact of the SLC unavailability. For a SLC unavailability lasting 4 hours (i.e. corresponding to the approximate time of the declared inoperability), the increase in conditional core damage probability (ICCDP) and large early release probability (ICLERP) are 9.4E-10 per year and 9.9E-10 per year, respectively. Increases in ICCDP and ICLERP values of less than 1.0E-06 per year and 1.0E-07 per year, respectively, are considered to be of very low safety significance. As a sensitivity to the duration of the unavailability, a SLC unavailability of approximately 17 days would have had to occur to exceed the ICCDP or ICLERP values associated with very low safety significance. Since the worst-case unavailability is bounded by 15 days due to the known satisfactory concentration on October 13, the event is below the threshold of very low safety significance. In addition to the PRA results, a deterministic evaluation of this function was performed. The minimum boron concentration in TS SR 3.1.7.5 is based on a DTE

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calculation which was performed to meet the Anticipated Transient Without Scram (ATWS) rule. The calculation contains an equivalency formula, consistent with an NRC-approved approach, for determining the sodium pentaborate concentration, B-10 enrichment, and SLC pump flow rate required to meet the rule. These three parameters are all related in that an increase in one parameter can offset a decrease in another parameter. Using this equivalency formula, ATWS rule compliance can be demonstrated for the sodium pentaborate concentration of 8.3% measured during the event even assuming the minimum B-10 enrichment of 65% in TS SR 3.1.7.10 and the minimum SLC pump flow rate of 41.2 gallons per minute (gpm) in TS SR 3.1.7.7. Therefore, although the SLC system was declared inoperable due to the low sodium pentaborate concentration, the actual SLC system parameters at the time of event would have ensured the capability to control reactivity in the unlikely event that the control rods could not be inserted. Note also that, during this event, the normal means of reactivity control (i.e. the control rods) was maintained.

The post-LOCA function of SLC to control pH is not modeled in the PRA. Therefore, only a deterministic approach was used to determine the safety consequences of the SLC system inoperability related to this function. The DTE calculation for pH assumes a certain minimum mass of sodium pentaborate available for injection. The minimum mass is based on the minimum tank volume and minimum sodium pentaborate concentration allowed by Figure 3.1.7-1 of TS 3.1.7. Although the sodium pentaborate concentration of 8.3% during the event was below the minimum value assumed in the calculation, the volume of solution in the tank at the time was much greater than that assumed in the calculation (i.e. near the high level alarm setpoint). A sensitivity calculation using the approximate volume and concentration at the time of the event results in an available mass of boron that is greater than that assumed in the DTE calculation for post-LOCA pH. The post-LOCA pH calculation assumes the manual injection of SLC is completed within 6 hours after the start of the LOCA. Using the approximate volume at the time of the event and the TS SR 3.1.7.7 minimum flow rate, the available mass of boron could be injected in approximately 75 minutes. Based on this sensitivity, although the SLC system was declared inoperable at the time of the event, the SLC system would have been capable of providing the sodium pentaborate mass required to ensure that post-LOCA pH control was successful.

Fermi 2 also has procedural guidance in place for an alternate means of injecting boron using the Standby Feedwater (SBFW) [[SJ]] system. The procedure requires that the SBFW system be capable of supplying water to the RPV and there be adequate water in the Condensate Storage Tank (CST) [[SD]] [[TK]]. Both of these requirements were met during the event such that alternate means of boron injection would have been available.

Based on the discussion above, the safety significance of this event is very low and the event did not pose a threat to the health and safety of the public or plant personnel. There were no radiological releases associated with this event.

CAUSE OF THE EVENT

The cause of the SLC storage tank sodium pentaborate concentration being below TS SR 3.1.7.5 was leak-by of the SLC storage tank demineralized water isolation valves while in their closed position. The increase in SLC storage tank level due to the leak-by of demineralized water diluted the sodium pentaborate solution in the tank. The cause of the unexpected increase in tank level and corresponding decrease in sodium pentaborate concentration going undetected was an inadequate system monitoring plan that did not adequately trend these parameters.

Failed components: SLC suction line and SLC storage tank demineralized water supply valves
Model number: 3624 (F316) CT, globe valve, 1-inch, stop valve assembly
Manufacturer: Rockwell-Edwards Univalve

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
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NARRATIVE

CORRECTIVE ACTIONS

Compliance with the TS requirement for sodium pentaborate concentration was restored by the addition of sodium pentaborate to the SLC storage tank once it was determined that the concentration did not meet the TS SR 3.1.7.5 limit. The two demineralized water isolation valves were replaced on November 9, 2016 to prevent further leak-by. The chemistry specification for the SLC storage tank was revised on November 24, 2016 to clearly specify action levels for the tank level and sodium pentaborate concentration. The action level for sodium pentaborate concentration is now 9.0%, which is the midpoint of the allowable concentration range in TS Figure 3.1.7-1 to provide additional margin to the TS SR. The SLC system monitoring plan was revised on December 5, 2016 to incorporate trending tank level and sodium pentaborate concentration to prevent a future undetected increase in level and decrease in boron concentration. Department human performance (HU) clock resets were conducted for Chemistry and Operations personnel to communicate lessons learned associated with this event.

Although not related to the increase in tank level and corresponding decrease in sodium pentaborate concentration, this event identified a small discrepancy between the MCR and local level indications that resulted in an alarm and the actual measured SLC storage tank level that was below the alarm setpoint as described in the narrative. The discrepancy was determined to be related to calibration. Re-calibration was performed on November 1, 2016 to resolve this issue.

PREVIOUS OCCURRENCES

A review of previous Fermi 2 LERs did not identify any reportable conditions affecting both SLC subsystems simultaneously. A review of the corrective action program did identify previous occurrences where an increase in SLC storage tank level was identified as resulting in a decrease in sodium pentaborate concentration. However, these occurrences were more than five years ago and none of these occurrences resulted in changes in level and concentration that were significant enough to exceed the TS SR. The corrective actions for one of these events did involve replacing a valve to eliminate potential leak-by of demineralized water into the SLC storage tank, similar to as discussed for this event. However, since these previous occurrences did not indicate a significant decrease in boron concentration, these past occurrences did not indicate a need to immediately replace valves or perform additional trending and therefore would not have been able to prevent the occurrence of this event.